

**25MT1306E - Mathematics for Data Science and Analytics**

**Project Report**

**on**

**Vehicle Price Analysis**

**Using Mathematical and Statistical Methods**

**Under the Guidance of**

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## BONAFIDE CERTIFICATE

This is to certify that the project titled "Vehicle Price Analysis Using Mathematical and Statistical Methods" is a bonafide record of the work done by

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Project Guide

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## ACKNOWLEDGEMENT

We express our sincere gratitude to the Department of Mathematics for providing us with the opportunity to complete this project as a part of the Mathematics for Data Science and Analytics course. This project helped us understand how mathematical and statistical concepts can be applied to a real-world vehicle pricing problem rather than remaining only as theoretical classroom topics.

We are deeply thankful to our faculty members, Dr. M P Mallesh and Mr. M. Nagesh, for their constant guidance, valuable suggestions, and encouragement throughout the project. Their academic support helped us understand the practical application of mathematical, statistical, and data analytical concepts in solving real-world problems. Their guidance helped us organize the work into a proper analytical sequence consisting of data cleaning, exploratory data analysis, probability analysis, statistical inference, regression modelling, and performance visualization.

We also thank K L Deemed to be University and the Department of Mathematics for providing the academic environment and resources required to complete this project. The project improved our practical understanding of Python libraries such as Pandas, NumPy, Matplotlib, Seaborn, SciPy, and Scikit-learn.

We extend our appreciation to all faculty members of the Department of Mathematics whose teaching and academic support throughout the course provided a strong foundation for this project. The concepts learned during classroom sessions played an important role in understanding data analysis, statistical reasoning, and predictive modelling techniques used in this study.

We acknowledge the open-source learning community and dataset providers for making structured data and documentation available. Working on this project helped us learn how raw data can be transformed into useful insights through systematic preprocessing, visualization, and model evaluation.

Finally, we thank all team members for their contribution, cooperation, and effort in completing the project successfully. Each stage of the project strengthened our understanding of data-driven decision making and predictive modelling.

## ABSTRACT

This project presents a comprehensive mathematical and statistical analysis of a used vehicle dataset. The dataset contains vehicle-related attributes such as vehicle name, year of manufacture, selling price, kilometers driven, fuel type, seller type, transmission, ownership status, mileage, engine capacity, maximum power, torque, and seating capacity. After data cleaning and preprocessing, the final dataset contained 6,926 vehicle records.

The primary objective of this study is to analyze the factors influencing used vehicle selling prices and to develop a predictive regression model capable of estimating vehicle resale value. The analytical pipeline includes data cleaning, duplicate removal, missing value imputation, data type conversion, exploratory data analysis, probability analysis, statistical inference, and multiple linear regression modelling.

Exploratory Data Analysis (EDA) was performed to identify patterns, trends, and relationships among variables. Statistical techniques such as probability analysis, confidence interval estimation, hypothesis testing, and regression analysis were applied to derive meaningful insights from the dataset. Feature engineering was also performed to improve model performance and predictive capability.

The results indicate that selling price is significantly influenced by vehicle age, maximum power, engine capacity, mileage, and ownership history. By extracting vehicle brand and model information and applying appropriate preprocessing techniques, the final regression model achieved an  $R^2$  score of approximately 0.804, demonstrating strong predictive performance. This project highlights the practical application of mathematical, statistical, and machine learning concepts in solving real-world data analytics problems and supporting data-driven decision making.

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# 1. Introduction

## 1.1 Background of the Topic

The used-vehicle market is an important part of the automobile industry because it allows buyers to purchase vehicles at lower prices and enables sellers to recover value from vehicles they no longer need. However, deciding a fair resale price is difficult because the value of a vehicle depends on many interrelated factors. These include the year of manufacture, kilometers driven, fuel type, transmission type, ownership history, engine capacity, maximum power, brand, and vehicle model.

Mathematical and statistical methods provide a structured way to analyze such price-related data. Instead of relying only on guesswork, historical records can be used to identify patterns and build models that estimate selling price. This project demonstrates how data science techniques can convert raw vehicle data into meaningful insights and predictive models.

The dataset is suitable for this analysis because it contains both numerical and categorical attributes. Numerical variables such as selling price, kilometers driven, mileage, engine capacity, and maximum power help quantify vehicle specifications, while categorical variables such as fuel type, seller type, transmission, owner category, brand, and vehicle model capture market identity and buyer preference.

## 1.2 Importance and Significance

Accurate vehicle price estimation is useful for buyers, sellers, dealerships, and online automobile platforms. Buyers can avoid overpaying, sellers can decide realistic asking prices, and platforms can recommend fair prices automatically. Statistical analysis also helps identify which features affect vehicle prices most strongly.

This project is significant because it connects course concepts such as probability, descriptive statistics, confidence intervals, hypothesis testing, correlation, and regression with a real-world pricing problem. The project also demonstrates how feature engineering can improve model performance beyond basic numerical analysis.

The analytical framework adopted in this project is based on established concepts from mathematical statistics, data analysis, regression modelling, and statistical learning. The methodologies used are consistent with standard approaches discussed in data science, statistical learning, and data mining literature, while modern Python-based analytical tools were employed to implement and evaluate these techniques using real-world vehicle data.

### Significance of the Study

The study shows that vehicle price prediction improves when raw variables are combined with engineered features such as brand and detailed vehicle model. This makes the analysis more realistic because market identity strongly influences resale value.

## 1.3 Dataset Overview

| Attribute Category | Examples | Purpose |
|--------------------|----------|---------|
|--------------------|----------|---------|

|                   |                                           |                                                             |
|-------------------|-------------------------------------------|-------------------------------------------------------------|
| Vehicle Identity  | name, brand, car_model                    | Captures manufacturer and variant-level pricing information |
| Price and Usage   | selling_price, km_driven, year            | Measures resale value, vehicle age, and usage               |
| Specifications    | mileage, engine, max_power, torque, seats | Represents performance and physical characteristics         |
| Market Categories | fuel, seller_type, transmission, owner    | Captures buyer preference and ownership conditions          |

After cleaning, the dataset contained 6,926 records. The final analysis focused on the selling price as the target variable. Important predictor variables included year, kilometers driven, mileage, engine capacity, maximum power, seats, fuel type, transmission type, seller type, ownership category, brand, and detailed vehicle model.

## 1.4 Problem Statement

Can we analyze the used-vehicle dataset to identify the major factors affecting selling price and build a regression model capable of predicting vehicle resale value with reasonable accuracy?

## 1.5 Objectives

| No. | Objective                                                                                             |
|-----|-------------------------------------------------------------------------------------------------------|
| 1   | Clean and preprocess the dataset by handling missing values, duplicates, and inconsistent data types. |
| 2   | Perform exploratory analysis to understand distributions, outliers, and relationships.                |
| 3   | Apply probability analysis to quantify the likelihood of important vehicle characteristics.           |
| 4   | Use statistical inference to estimate and test meaningful price-related conclusions.                  |
| 5   | Build and compare multiple regression models for vehicle price prediction.                            |
| 6   | Improve model performance using feature engineering and summarize results using a dashboard.          |

## 1.6 Tools and Libraries

| Library | Purpose                                                    |
|---------|------------------------------------------------------------|
| Python  | Core programming language used for analysis and modeling   |
| Pandas  | Data loading, cleaning, manipulation, and tabular analysis |
| NumPy   | Numerical computation and array operations                 |

|              |                                                                   |
|--------------|-------------------------------------------------------------------|
| Matplotlib   | Base plotting library for charts and visualizations               |
| Seaborn      | Statistical visualization including heatmaps and regression plots |
| SciPy        | Statistical inference including confidence interval and t-test    |
| Scikit-learn | Train-test split, linear regression, and model evaluation metrics |



## 2. Data Cleaning and Preprocessing - M1

Data cleaning is the foundation of any data science project. Raw datasets often contain missing values, duplicate records, inconsistent data types, and outliers. If these issues are not handled properly, later stages such as EDA, statistical inference, and regression modeling may produce unreliable results. In this project, the cleaning stage focused on preparing the vehicle dataset for mathematical and statistical analysis.

### 2.1 Loading the Dataset

The dataset was loaded using Pandas. After loading, the shape, first few rows, column names, and data types were inspected. This helped identify which variables were numerical and which variables were categorical.

```
# Step 1: Read the data
df = pd.read_csv('Vehicle details v3.csv')

# Step 2: Check shape and preview
print(df.shape)
print(df.head())
print(df.info())
```

The initial dataset contained more than 8,000 records before duplicate removal. After cleaning and removing repeated rows, 6,926 valid records were retained for the final analysis.

### 2.2 Handling Missing Values

Missing values were checked column-wise using `isnull().sum()`. Numerical columns were filled using the median because the median is less affected by extreme values. Categorical columns were filled using the mode because it represents the most frequent category.

```
# Fill numerical columns with median
num_cols = df.select_dtypes(include='number').columns
df[num_cols] = df[num_cols].fillna(df[num_cols].median())

# Fill categorical columns with mode
cat_cols = df.select_dtypes(include='object').columns
for col in cat_cols:
    df[col] = df[col].fillna(df[col].mode()[0])
```

This ensured that no missing values remained before the dataset was used for analysis and modeling.

### 2.3 Removing Duplicate Records

Duplicate records can distort analysis by giving repeated observations extra weight. The dataset contained 1,202 duplicate records. These were removed using `drop_duplicates()`, and the index was reset.

```
# Check and remove duplicates
print(df.duplicated().sum())
df.drop_duplicates(inplace=True)
df.reset_index(drop=True, inplace=True)
print(df.duplicated().sum())
```

After duplicate removal, the dataset became more reliable and represented unique vehicle records only.

## 2.4 Data Type Conversion

Some important numerical variables were stored as object/string values because they contained units such as 'kmpl', 'CC', and 'bhp'. These unit strings were removed and the columns were converted into numerical format.

```
df['mileage'] = df['mileage'].str.replace(' kmpl', '')
df['mileage'] = pd.to_numeric(df['mileage'], errors='coerce')

df['engine'] = df['engine'].str.replace(' CC', '')
df['engine'] = pd.to_numeric(df['engine'], errors='coerce')

df['max_power'] = df['max_power'].str.replace(' bhp', '')
df['max_power'] = pd.to_numeric(df['max_power'], errors='coerce')
```

This conversion was essential because mathematical calculations, correlation analysis, and regression modeling require numeric input values.

## 2.5 Outlier Detection

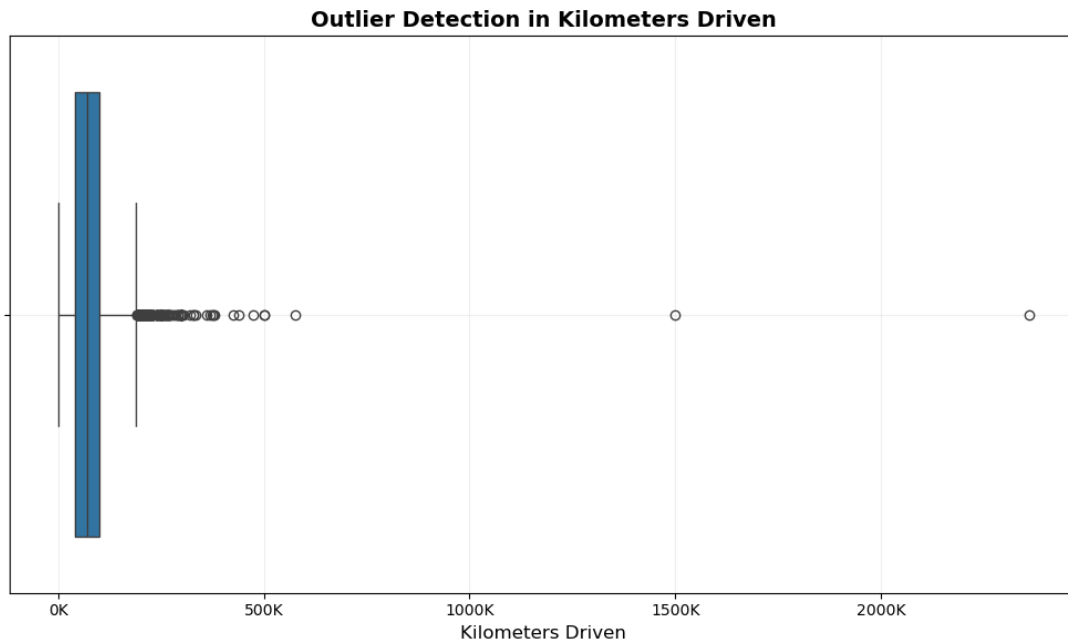
Outliers were studied using boxplots. Outliers are not always errors; in vehicle pricing, they may represent luxury cars, unusually high mileage vehicles, or high-performance vehicles. Therefore, the project identifies and interprets outliers instead of blindly deleting them.



**Figure 2.1: Outlier Detection in Selling Price**

**Diagram Description:** This boxplot shows the spread of selling prices in the cleaned dataset. The majority of vehicles are concentrated in the lower price range, while several points lie far to the right of the whisker.

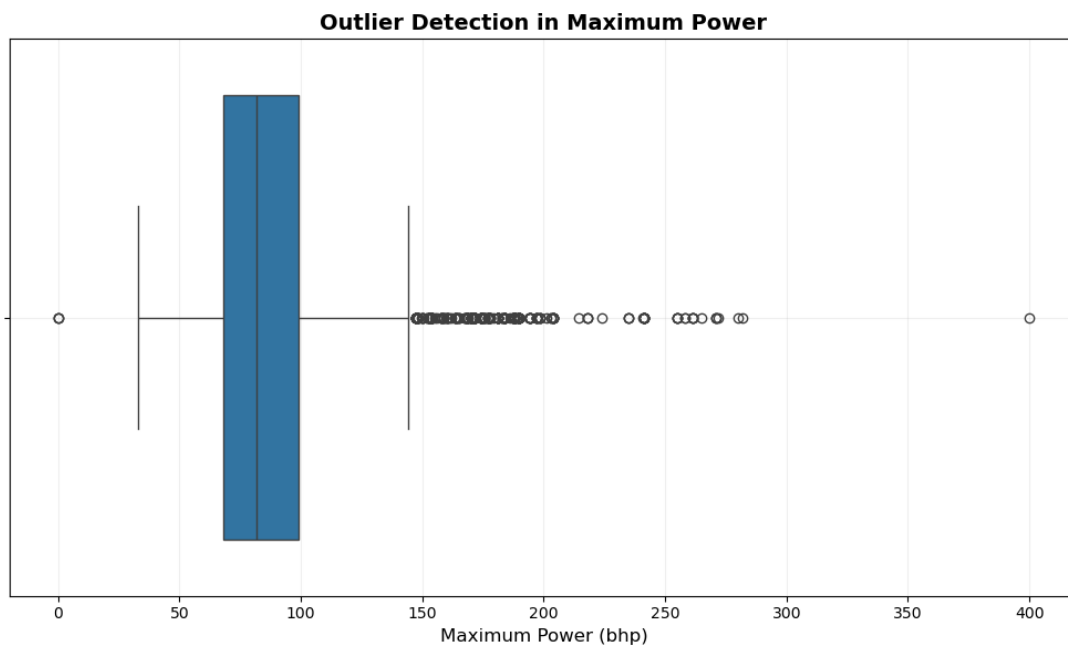
**Interpretation:** The extreme values represent premium or luxury vehicles with unusually high resale prices. These values explain why the selling price distribution is positively skewed and why median-based methods are useful during cleaning.



**Figure 2.2: Outlier Detection in Kilometers Driven**

**Diagram Description:** This boxplot displays the distribution of kilometers driven. Most vehicles fall within a moderate usage range, while some records show very high kilometer readings.

**Interpretation:** High kilometer values may represent older commercial or heavily used vehicles. These values are important because usage directly affects resale value and buyer preference.



**Figure 2.3: Outlier Detection in Maximum Power**

**Diagram Description:** This boxplot shows the distribution of maximum power values measured in bhp. A few high-power vehicles appear as outliers.

**Interpretation:** High maximum power values usually correspond to premium or performance-oriented vehicles. These outliers are meaningful market observations rather than simple data errors.

## 2.6 M1 Summary

| Cleaning Step        | Result                                                               |
|----------------------|----------------------------------------------------------------------|
| Missing values       | Handled using median and mode imputation                             |
| Duplicates           | 1,202 duplicate records removed                                      |
| Data type conversion | mileage, engine, and max_power converted to numeric                  |
| Outlier analysis     | Performed using boxplots for selling_price, km_driven, and max_power |
| Final dataset        | 6,926 cleaned records                                                |

### M1 Observation

The dataset became clean, consistent, and ready for further analysis. The presence of outliers was expected in a used-vehicle dataset because premium vehicles, high-mileage vehicles, and powerful vehicles naturally create extreme values.

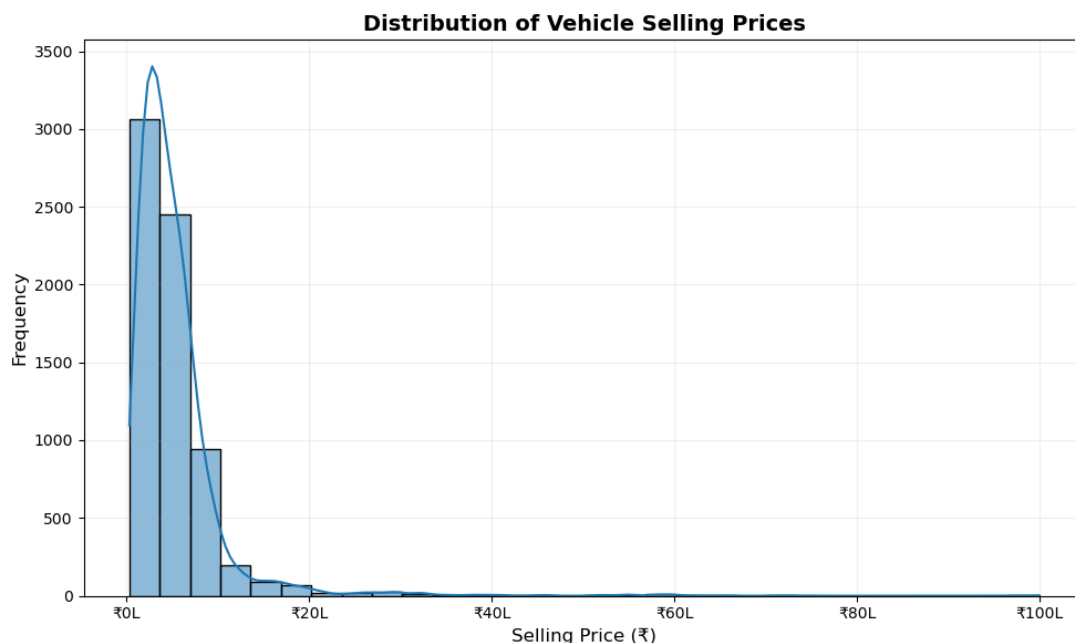
### 3. Exploratory Data Analysis - M2

Exploratory Data Analysis (EDA) is used to understand the structure, distribution, and relationships within the dataset. In this project, EDA helped identify price patterns, category distributions, feature relationships, and potential predictors of selling price.

#### 3.1 EDA Summary

| Figure | Visualization              | Key Insight                                             |
|--------|----------------------------|---------------------------------------------------------|
| 3.1    | Selling Price Histogram    | Selling price is positively skewed                      |
| 3.2    | Fuel Type Bar Chart        | Diesel and Petrol vehicles dominate                     |
| 3.3    | Transmission Bar Chart     | Manual transmission is most common                      |
| 3.4    | Correlation Heatmap        | max_power has strongest positive correlation with price |
| 3.5    | Selling Price vs Max Power | Higher power generally increases selling price          |

#### 3.2 Selling Price Distribution

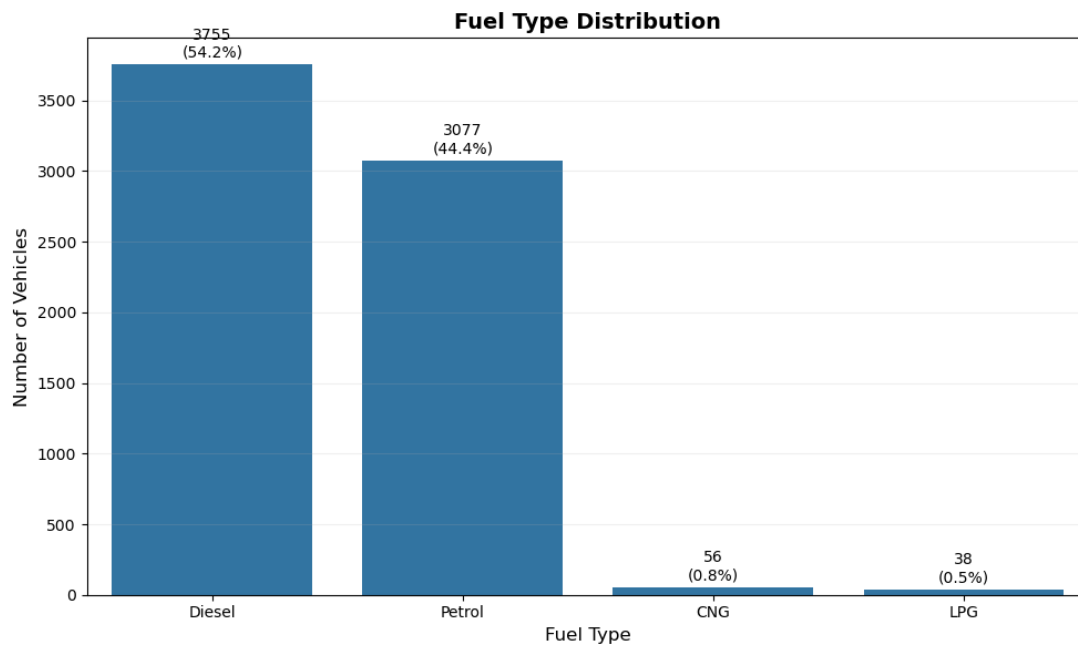


**Figure 3.1: Distribution of Vehicle Selling Prices**

**Diagram Description:** The histogram shows the distribution of selling prices after cleaning. Most vehicles fall in the lower and middle price ranges, while a smaller number of expensive vehicles extends the distribution to the right.

**Interpretation:** The distribution is right-skewed. The mean selling price is higher than the median, indicating that a few high-value vehicles increase the average price. This supports the use of median-based imputation during cleaning.

### 3.3 Fuel Type Distribution

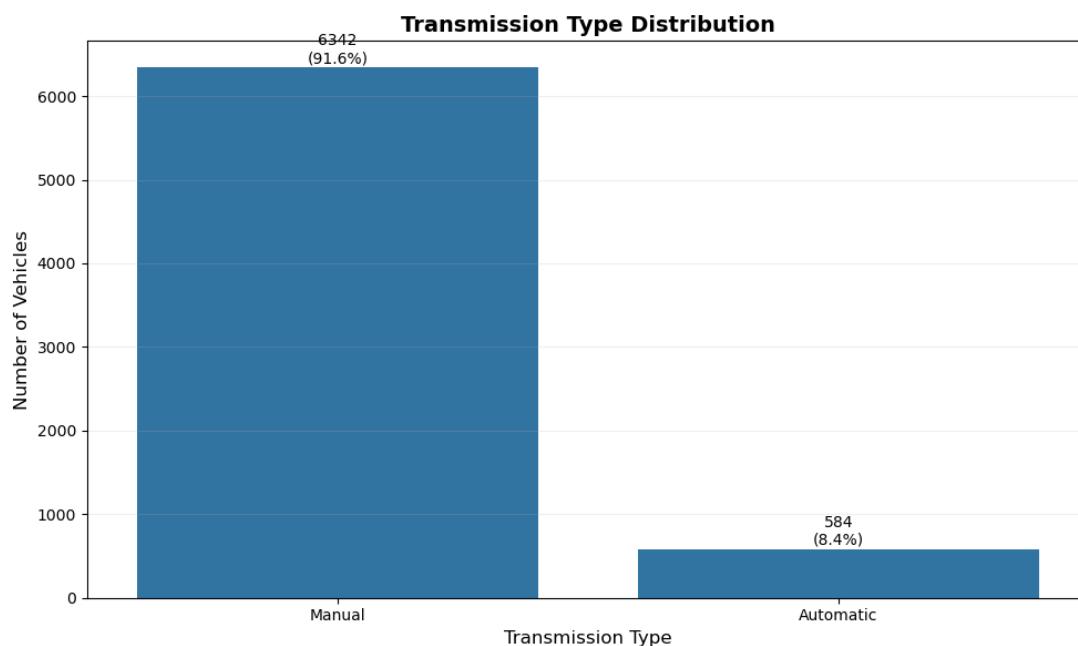


**Figure 3.2: Fuel Type Distribution**

Diagram Description: This bar chart shows the number and percentage of vehicles belonging to each fuel type category. Diesel and Petrol vehicles form the majority of the dataset.

Interpretation: Diesel vehicles slightly outnumber Petrol vehicles. CNG and LPG vehicles appear in much smaller proportions, showing that the used-vehicle market in this dataset is dominated by Diesel and Petrol vehicles.

### 3.4 Transmission Distribution

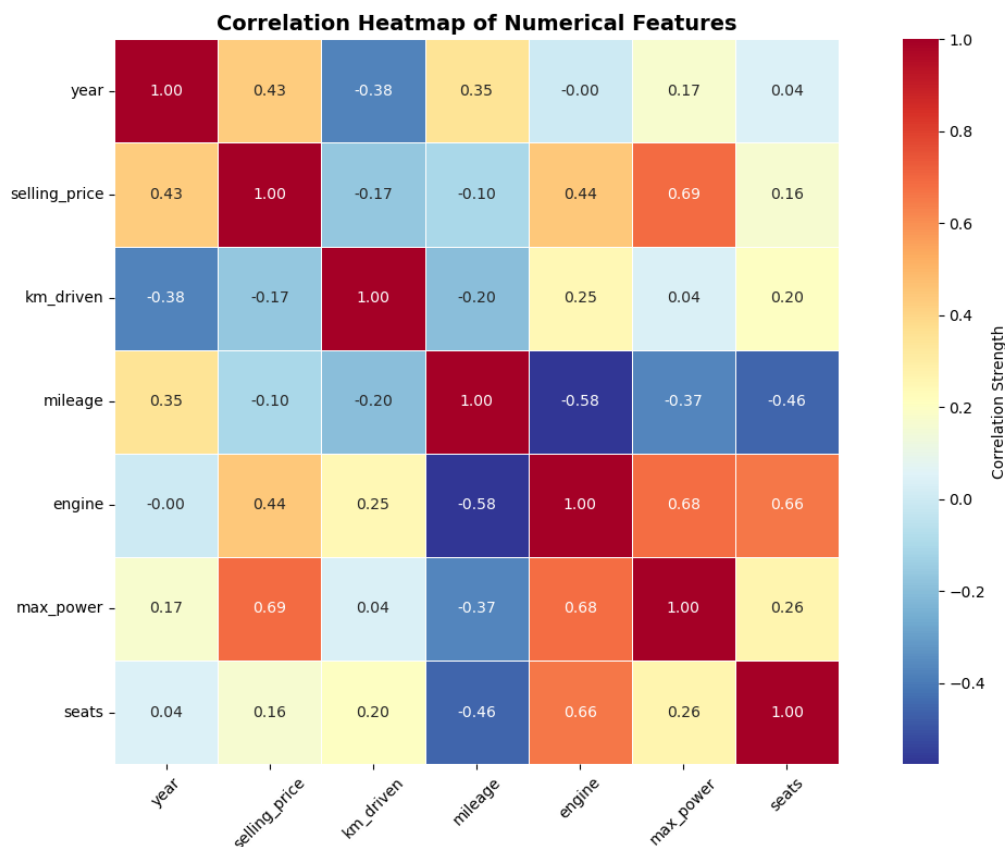


**Figure 3.3: Transmission Type Distribution**

Diagram Description: This chart compares the number of Manual and Automatic vehicles. Manual vehicles occur much more frequently than Automatic vehicles.

Interpretation: Manual transmission dominates the dataset with more than 91% probability of occurrence. This reflects common market conditions where manual vehicles are more common in used-vehicle listings.

### 3.5 Correlation Heatmap

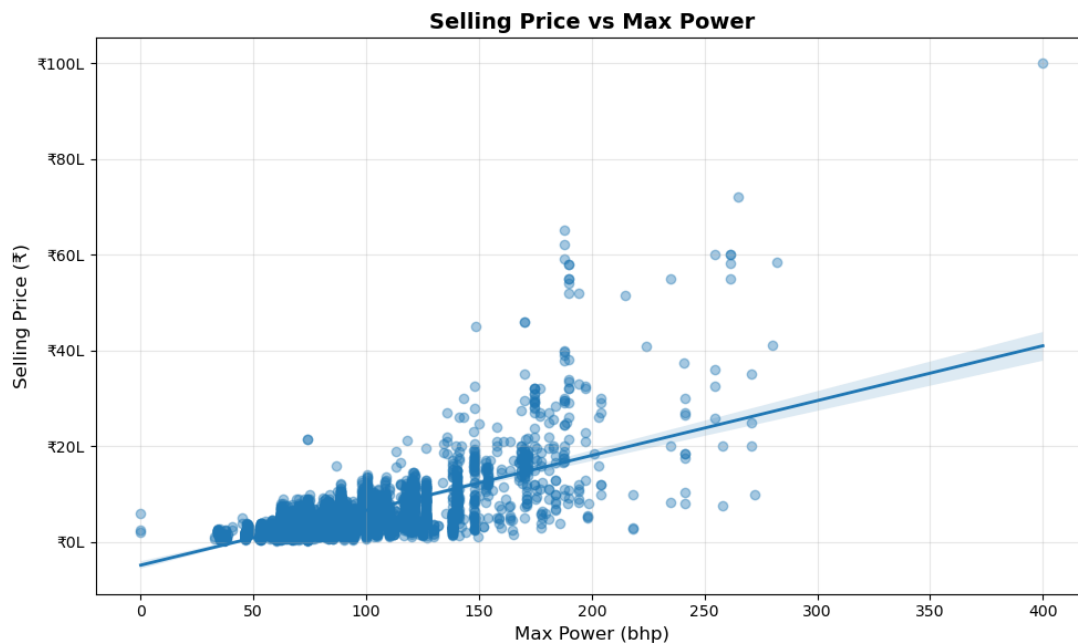


**Figure 3.4: Correlation Heatmap of Numerical Features**

Diagram Description: The heatmap shows Pearson correlation coefficients among numerical variables. Positive values indicate direct relationships and negative values indicate inverse relationships.

Interpretation: max\_power shows the strongest positive correlation with selling price. Engine capacity and year also show moderate positive correlations, while kilometers driven has a weak negative relationship with selling price.

### 3.6 Selling Price vs Maximum Power



**Figure 3.5: Selling Price vs Maximum Power**

**Diagram Description:** The scatter plot compares maximum power against selling price. The trend line shows the overall direction of the relationship.

**Interpretation:** Vehicles with higher maximum power generally have higher selling prices. However, the spread of points shows that power alone is not enough to fully explain price variation; brand, model, year, and other factors also matter.

### 3.7 M2 Summary

The EDA stage revealed that the dataset is dominated by Diesel/Petrol vehicles and Manual transmissions. Selling price is positively skewed due to premium vehicles. Among numerical features, max\_power, engine capacity, and year are the strongest predictors of selling price. EDA therefore guided feature selection for the regression modeling stage.

#### **M2 Observation**

The EDA results confirm that vehicle performance, age, and market identity influence used-vehicle selling prices. These findings motivated the inclusion of both numerical and categorical variables in later models.



## 4. Probability Analysis - M3

Probability analysis quantifies the likelihood of different vehicle characteristics occurring in the dataset. This helps transform categorical observations into measurable insights. In this project, probability was calculated for fuel type, transmission, ownership status, and seller type.

### 4.1 Basic Probability

The probability of an event is calculated as the number of favorable outcomes divided by the total number of observations. For example, the probability of selecting a Diesel vehicle is calculated as the number of Diesel vehicles divided by the total number of vehicles.

```
diesel_count = len(df[df['fuel'] == 'Diesel'])
total_count = len(df)
prob_diesel = diesel_count / total_count

manual_count = len(df[df['transmission'] == 'Manual'])
prob_manual = manual_count / total_count
```

| Characteristic      | Probability |
|---------------------|-------------|
| Diesel vehicle      | 54.22%      |
| Petrol vehicle      | 44.42%      |
| Manual transmission | 91.57%      |
| First owner vehicle | 61.25%      |
| Individual seller   | 89.78%      |

### 4.2 Probability Visualization

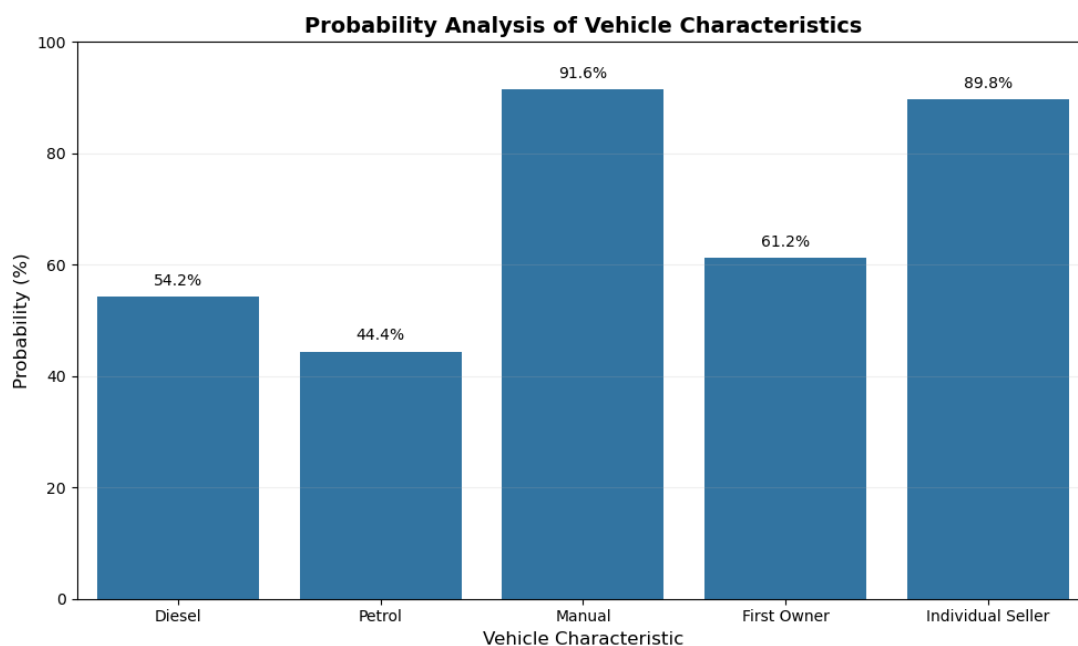


Figure 4.1: Probability Analysis of Vehicle Characteristics

Diagram Description: The bar chart compares the probability of major vehicle characteristics such as Diesel fuel, Petrol fuel, Manual transmission, First Owner vehicles, and Individual Seller listings.

Interpretation: Manual transmission and Individual Seller categories have the highest probabilities. This confirms that the dataset is heavily dominated by manually driven vehicles sold by individual sellers.

### 4.3 Interpretation of Probability Results

Probability analysis helps understand the expected composition of the dataset. For example, if one vehicle is randomly selected from the cleaned dataset, it is highly likely to be a Manual transmission vehicle and also likely to be sold by an Individual Seller. Such probability values are useful for understanding dataset imbalance and market representation.

The probability of selecting a First Owner vehicle is also high, indicating that many listings are relatively early in ownership history. Since ownership status can influence resale value, it becomes an important categorical variable for regression modeling.

### 4.4 M3 Summary

#### M3 Observation

Probability analysis showed that Manual transmission and Individual Seller vehicles dominate the dataset. Diesel vehicles are slightly more common than Petrol vehicles, and First Owner vehicles form a large portion of the records.

## 5. Statistical Inference - M4

Statistical inference allows conclusions to be drawn about the wider population using sample data. In this project, inference was used to estimate the average selling price and test whether the observed mean selling price differs significantly from a hypothesized value.

### 5.1 Mean Selling Price

The mean selling price of vehicles in the cleaned dataset was approximately Rs. 5.17 lakh. Since the dataset contains vehicles from many price ranges, including luxury vehicles, the mean is influenced by high-price outliers.

| Metric                              | Value        |
|-------------------------------------|--------------|
| Mean Selling Price                  | Rs. 5,17,271 |
| 95% Confidence Interval Lower Bound | Rs. 5,05,029 |
| 95% Confidence Interval Upper Bound | Rs. 5,29,512 |
| t-statistic                         | 2.7653       |
| p-value                             | 0.0057       |

### 5.2 Confidence Interval

A 95% confidence interval was calculated for the population mean selling price. The confidence interval ranged from approximately Rs. 5.05 lakh to Rs. 5.30 lakh. This means that based on the sample data, the true average used-vehicle selling price is expected to lie within this interval with 95% confidence.

```
mean_price = df['selling_price'].mean()
std_price = df['selling_price'].std()
n = len(df)
confidence_level = 0.95
t_value = stats.t.ppf((1 + confidence_level) / 2, n-1)
margin_error = t_value * (std_price / np.sqrt(n))
lower_bound = mean_price - margin_error
upper_bound = mean_price + margin_error
```

### 5.3 One-Sample t-Test

A one-sample t-test was performed using Rs. 5,00,000 as the hypothesized mean selling price. The null hypothesis assumed that the average selling price equals Rs. 5,00,000. The p-value obtained was 0.0057, which is less than 0.05.

Since the p-value is below the 5% significance level, the null hypothesis is rejected. This indicates that the average selling price in the dataset is statistically different from Rs. 5,00,000.

## 5.4 M4 Summary

### M4 Observation

The statistical inference section confirmed that the mean selling price is approximately Rs. 5.17 lakh and is significantly different from the hypothesized value of Rs. 5,00,000. The confidence interval supports the reliability of this conclusion.

## 6. Predictive Modeling and Regression Analysis - M5

Predictive modeling was performed to estimate vehicle selling prices using Linear Regression. The target variable was `selling_price`, while predictor variables included numerical features, categorical features, and engineered features extracted from the vehicle name.

### 6.1 Regression Modeling Approach

Multiple regression models were developed using different combinations of features. The purpose was to observe how prediction performance changes as more relevant variables are included. The models were evaluated using MAE, MSE, RMSE, and  $R^2$  Score.

| Metric      | Meaning                                              |
|-------------|------------------------------------------------------|
| MAE         | Average absolute prediction error                    |
| MSE         | Average squared prediction error                     |
| RMSE        | Error measure in the original price scale            |
| $R^2$ Score | Proportion of price variation explained by the model |

### 6.2 Feature Engineering

A major improvement in model performance came from feature engineering. Initially, the model used basic numerical variables such as engine capacity, maximum power, year, kilometers driven, mileage, and seats. However, used-vehicle prices depend strongly on vehicle identity. A budget hatchback, a mid-range sedan, and a premium SUV may have very different market values even when some numerical features overlap.

To capture this hidden information, the vehicle name column was used to extract brand and detailed vehicle model/variant information. These categorical values were then encoded using one-hot encoding and included in the regression model.

```
df_model['brand'] = df_model['name'].str.split().str[0]
df_model['car_model'] = (
    df_model['name']
    .str.split()
    .str[1:4]
    .str.join('_')
)

df_model = pd.get_dummies(
    df_model,
    columns=['fuel', 'seller_type', 'transmission', 'owner', 'brand',
'car_model'],
    drop_first=True
)
```

#### Why Feature Engineering Worked

The name column contained important market information. By extracting brand and detailed vehicle model/variant, the model learned manufacturer-specific and model-specific pricing patterns that were not present in basic numerical variables alone.

### 6.3 Model Comparison

| Rank | Model   | Main Features                                   | R <sup>2</sup> Score |
|------|---------|-------------------------------------------------|----------------------|
| 1    | Model 8 | Numerical + categorical + brand + vehicle model | 0.804                |
| 2    | Model 6 | Numerical + fuel + transmission                 | 0.563                |
| 3    | Model 3 | Engine + max_power + year                       | 0.562                |
| 4    | Model 7 | Seller and owner added                          | 0.547                |
| 5    | Model 5 | Numerical + mileage + seats                     | 0.546                |
| 6    | Model 4 | Top numerical features                          | 0.545                |
| 7    | Model 1 | Max power only                                  | 0.436                |
| 8    | Model 2 | Engine + max power                              | 0.435                |

### 6.4 R<sup>2</sup> = 0.804 Discussion

The final model achieved an R<sup>2</sup> score of approximately 0.804. This means that about 80.4% of the variation in vehicle selling price was explained by the selected features. The improvement was not random; it came from systematic model comparison and feature engineering.

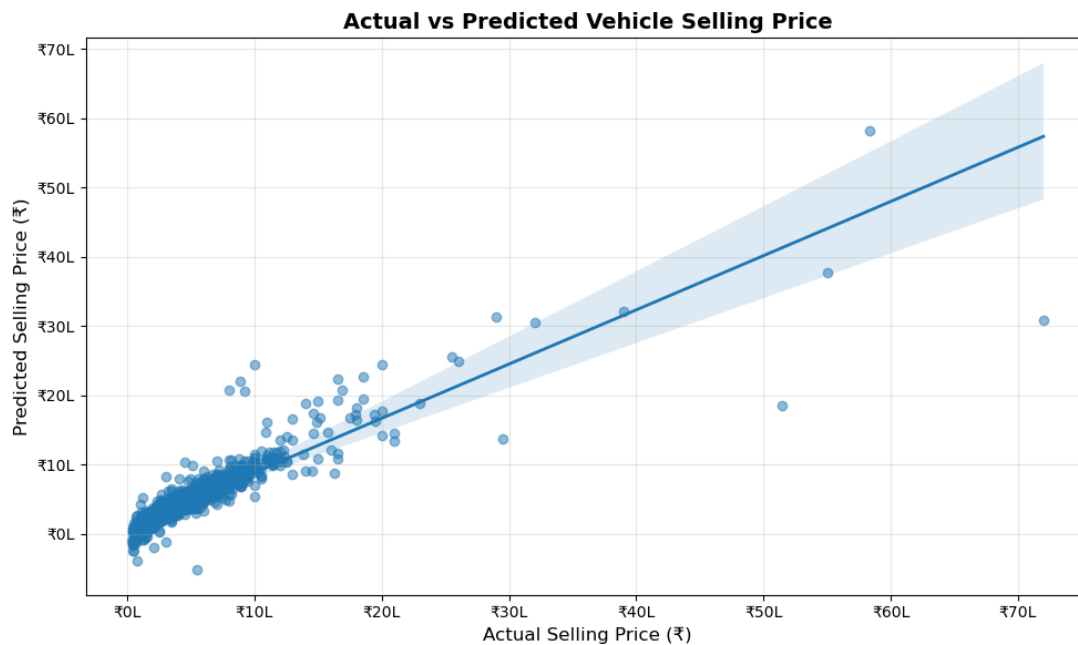
The initial full numerical/categorical model gave moderate performance. After adding brand information, performance increased significantly because different manufacturers have different resale values. After adding detailed vehicle model and variant information, the model improved further because model-level identity captures features such as segment, popularity, market demand, and specification differences.

The model did not reach 100% because used-vehicle prices also depend on factors not present in the dataset, such as vehicle condition, accident history, service records, insurance status, exact location, negotiation, modifications, and current market demand.

### 6.5 Model Evaluation Summary

| Final Metric         | Value        |
|----------------------|--------------|
| Best Model           | Model 8      |
| R <sup>2</sup> Score | 0.804        |
| RMSE                 | Rs. 2,07,320 |
| MAE                  | Rs. 87,969   |
| Total Features Used  | 1,383        |

## 6.6 Actual vs Predicted Visualization



**Figure 6.1: Actual vs Predicted Vehicle Selling Price**

**Diagram Description:** This regression plot compares actual selling prices with predicted selling prices from the final model. The trend line shows the overall alignment between actual and predicted values.

**Interpretation:** The strong positive pattern indicates that the final regression model predicts vehicle prices reasonably well. Some spread remains, which is expected because real-world vehicle prices are affected by additional factors not present in the dataset.

## 6.7 M5 Summary

### M5 Observation

Eight regression models were compared. Model 8 performed best after including numerical features, categorical features, brand information, and detailed vehicle model/variant features. The final  $R^2$  score of 0.804 shows strong predictive performance for a real-world used-vehicle dataset.

## 7. Model Visualization and Performance Dashboard - M6

The final module summarizes the performance of the selected model visually and numerically. The Actual vs Predicted plot helps evaluate how closely predicted prices follow actual prices, while the dashboard presents the most important performance metrics in a compact format.

### 7.1 Dashboard Metrics

| Metric               | Value        |
|----------------------|--------------|
| Total Records        | 6,926        |
| Total Features Used  | 1,383        |
| Best Model           | Model 8      |
| R <sup>2</sup> Score | 0.804        |
| RMSE                 | Rs. 2,07,320 |
| MAE                  | Rs. 87,969   |

The dashboard confirms that the final model explains around 80% of price variation. RMSE and MAE show the average prediction errors in rupee terms, making the model results easier to interpret for real-world vehicle pricing.

### 7.2 Practical Interpretation

A mean absolute error of around Rs. 87,969 means that, on average, the model's predicted vehicle price differs from the actual selling price by about this amount. For a used-vehicle dataset containing both budget and premium vehicles, this is a reasonable error level.

The RMSE is higher than MAE because it gives more weight to large errors. Large errors are expected for luxury vehicles and unusual records because such prices depend on factors like vehicle condition, modifications, accident history, and exact variant details.

### 7.3 M6 Summary

#### M6 Observation

The dashboard and Actual vs Predicted plot confirm that the final model is effective for price estimation. The model is not perfect, but it provides a strong and explainable prediction system based on available vehicle characteristics.



## 8. Final Project Conclusion

This project successfully applied mathematical, statistical, and machine learning techniques to analyze a used vehicle dataset. The complete workflow included data cleaning, missing value treatment, duplicate removal, data type conversion, outlier analysis, exploratory data analysis, probability analysis, statistical inference, multiple regression modeling, and dashboard visualization.

Exploratory analysis revealed that selling prices are positively skewed and that most vehicles belong to the Diesel/Petrol and Manual transmission categories. The correlation heatmap showed that maximum power, engine capacity, and manufacturing year are important numerical predictors of selling price.

Probability analysis quantified the occurrence of important vehicle characteristics, while statistical inference provided confidence-based conclusions about the average selling price. The mean selling price was approximately Rs. 5.17 lakh, and the 95% confidence interval suggested that the true average used-vehicle selling price lies between Rs. 5.05 lakh and Rs. 5.30 lakh.

The strongest part of the project was predictive modeling. Multiple regression models were compared, and the best model achieved an  $R^2$  score of approximately 0.804. This improvement was mainly achieved through feature engineering by extracting brand and detailed vehicle model information from the name column.

Overall, the project demonstrates that data science techniques can effectively support used-vehicle price estimation. The final model is explainable, statistically meaningful, and practically useful for understanding the major factors affecting resale value.

### 8.1 Key Outcomes

| Outcome                                    | Result            |
|--------------------------------------------|-------------------|
| Cleaned records                            | 6,926             |
| Duplicate rows removed                     | 1,202             |
| Most common transmission                   | Manual            |
| Dominant fuel categories                   | Diesel and Petrol |
| Strongest numerical correlation with price | max_power         |
| Best model                                 | Model 8           |
| Final $R^2$ Score                          | 0.804             |

### 8.2 Limitations

Although the model performed well, some important real-world variables were not available in the dataset. These include accident history, service records, insurance status, physical condition, location-specific demand, number of previous repairs, and negotiation details. Including such features could further improve prediction accuracy.

### **8.3 Future Scope**

Future work can include advanced algorithms such as Random Forest Regressor, Gradient Boosting, XGBoost, and CatBoost. These models can capture non-linear relationships better than Linear Regression. A web-based price prediction application can also be built using the final trained model.

## References

The following references, datasets, documentation sources, and academic resources were consulted during the development of this project.

| No. | Reference Details                                                                                                                          |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------|
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| 2   | James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). An Introduction to Statistical Learning with Applications in Python. Springer. |
| 3   | Montgomery, D. C., Peck, E. A., & Vining, G. G. (2021). Introduction to Linear Regression Analysis (6th Edition). Wiley.                   |
| 4   | Gupta, S. C., & Kapoor, V. K. Fundamentals of Mathematical Statistics. Sultan Chand & Sons.                                                |
| 5   | Han, J., Kamber, M., & Pei, J. Data Mining: Concepts and Techniques (3rd Edition). Morgan Kaufmann.                                        |
| 6   | CarDekho Used Vehicle Dataset (Kaggle Dataset Repository).                                                                                 |
| 7   | Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830.               |
| 8   | Harris, C. R., et al. (2020). Array Programming with NumPy. Nature, 585(7825), 357–362.                                                    |
| 9   | Pandas Documentation. <a href="https://pandas.pydata.org/">https://pandas.pydata.org/</a>                                                  |
| 10  | Scikit-learn Documentation. <a href="https://scikit-learn.org/">https://scikit-learn.org/</a>                                              |

## Appendices

### Appendix A - Data Dictionary

This appendix summarizes the major variables used in the Vehicle Price Analysis project. The dataset contains both numerical and categorical attributes. Numerical variables are useful for mathematical calculations and regression analysis, while categorical variables help capture market segments and vehicle identity.

| Column        | Type        | Description                                                        |
|---------------|-------------|--------------------------------------------------------------------|
| name          | Categorical | Full vehicle name containing brand, model, and variant information |
| year          | Numerical   | Manufacturing year of the vehicle                                  |
| selling_price | Numerical   | Target variable representing resale price                          |
| km_driven     | Numerical   | Total kilometers driven by the vehicle                             |
| fuel          | Categorical | Fuel type such as Diesel, Petrol, CNG, or LPG                      |
| transmission  | Categorical | Manual or Automatic transmission                                   |
| owner         | Categorical | Ownership history of the vehicle                                   |
| mileage       | Numerical   | Fuel efficiency after unit cleaning                                |
| engine        | Numerical   | Engine capacity in CC after conversion                             |
| max_power     | Numerical   | Maximum power in bhp after conversion                              |

The name column became especially important in the modeling stage because it contained hidden information about brand and vehicle model. Extracting this information helped improve the final  $R^2$  score substantially.

### Appendix B - Data Cleaning Checklist

The data cleaning stage ensured that the dataset was suitable for statistical analysis and machine learning. Each cleaning step had a specific purpose and directly improved the reliability of the project.

| Cleaning Task  | Action Taken             | Reason                                      |
|----------------|--------------------------|---------------------------------------------|
| Missing values | Median/mode imputation   | Preserves data while reducing distortion    |
| Duplicate rows | 1,202 duplicates removed | Avoids repeated records influencing results |
| Text units     | Removed kmpl, CC, bhp    | Required for numerical analysis             |

|                     |                      |                                                   |
|---------------------|----------------------|---------------------------------------------------|
| Datatype conversion | Converted to numeric | Needed for correlation and regression             |
| Outlier study       | Boxplots used        | Identifies extreme but meaningful vehicle records |

The cleaning process was not limited to removing errors. In this dataset, some extreme values represented real market cases such as luxury vehicles or high-performance vehicles. Therefore, the project interpreted outliers carefully instead of blindly deleting them.

### Code Used for Final Dataset Validation

```
print("Final Shape:", df.shape)
print("Missing Values:", df.isnull().sum().sum())
print("Duplicate Records:", df.duplicated().sum())
print(df.dtypes)
```

## Appendix C - Detailed EDA Interpretation

Exploratory Data Analysis helped convert raw data into visual and statistical understanding. The selling price histogram showed right skewness, which is common in used-vehicle price datasets because most vehicles are affordable while a few premium vehicles create a long upper tail.

Fuel type and transmission distribution graphs revealed dataset imbalance. Diesel and Petrol vehicles dominated the fuel categories, while Manual transmission dominated the transmission category. These insights helped explain why categorical encoding was necessary before modeling.

The correlation heatmap showed that max\_power had the strongest positive relationship with selling\_price among the numerical variables. Engine capacity and year also showed useful positive relationships. This supported the selection of these variables in early regression models.

The Selling Price vs Max Power plot confirmed that powerful vehicles generally sell at higher prices. However, the scatter also showed variation, proving that power alone cannot explain resale price completely.

### EDA Code Pattern

```
plt.figure(figsize=(10,6))
sns.histplot(df['selling_price'], bins=30, kde=True)
plt.title('Distribution of Vehicle Selling Prices')
plt.xlabel('Selling Price')
plt.ylabel('Frequency')
plt.show()
```

## Appendix D - Probability Calculation Notes

Probability analysis was used to quantify the likelihood of different vehicle categories. The project calculated probabilities using the relative frequency method, where probability equals favorable outcomes divided by total observations.

Probability = Number of favorable outcomes / Total number of observations

Example:

$P(\text{Diesel}) = \text{Number of Diesel vehicles} / \text{Total number of vehicles}$

| Event          | Probability | Interpretation                                |
|----------------|-------------|-----------------------------------------------|
| Diesel vehicle | 54.22%      | Diesel vehicles are slightly more common than |

|                     |        |                                                      |
|---------------------|--------|------------------------------------------------------|
|                     |        | Petrol vehicles                                      |
| Petrol vehicle      | 44.42% | Petrol vehicles are the second-largest fuel category |
| Manual transmission | 91.57% | Manual vehicles dominate the dataset                 |
| First owner vehicle | 61.25% | Most vehicles belong to the first-owner category     |
| Individual seller   | 89.78% | Most vehicles are sold by individual sellers         |

These probabilities helped describe the composition of the dataset mathematically and supported later interpretations about market representation.

## Appendix E - Statistical Inference Notes

Statistical inference was used to draw conclusions about the broader vehicle population from the sample dataset. The confidence interval estimated the likely range of the true average selling price, while the one-sample t-test checked whether the observed mean differed significantly from a hypothesized value.

95% Confidence Interval:  
 $\text{mean} \pm t\text{-critical} * (\text{standard deviation} / \sqrt{n})$

One-sample t-test:  
 $H_0$ : mean selling price = Rs. 5,00,000  
 $H_1$ : mean selling price  $\neq$  Rs. 5,00,000

The p-value obtained from the t-test was 0.0057. Since this value is lower than 0.05, the null hypothesis was rejected. This means the average selling price in the dataset was statistically different from Rs. 5,00,000.

The 95% confidence interval ranged approximately from Rs. 5.05 lakh to Rs. 5.30 lakh, supporting the conclusion that the average price is close to Rs. 5.17 lakh.

## Appendix F - Regression Model Progression

The regression modeling section followed a gradual model-building strategy. Instead of building only one model, multiple models were compared to understand how different sets of features affected predictive performance.

| Stage     | Feature Set                        | Purpose                                    |
|-----------|------------------------------------|--------------------------------------------|
| Model 1   | max_power                          | Tests strongest single numerical predictor |
| Model 2   | engine + max_power                 | Adds engine capacity                       |
| Model 3   | engine + max_power + year          | Adds vehicle age-related information       |
| Model 4-7 | Numerical + categorical groups     | Tests progressive feature combinations     |
| Model 8   | Full model + brand + vehicle model | Final feature-engineered model             |

The final model performed best because it included both measurable vehicle specifications and vehicle identity features. This allowed the regression model to capture manufacturer and model-level effects.

### Evaluation Function

```
def evaluate_model(features):
    X = df_model[features]
    y = df_model['selling_price']
    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.2, random_state=42
    )
    model = LinearRegression()
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    mae = mean_absolute_error(y_test, y_pred)
    mse = mean_squared_error(y_test, y_pred)
    rmse = np.sqrt(mse)
    r2 = r2_score(y_test, y_pred)
    return mae, mse, rmse, r2
```

## Appendix G - Feature Engineering Explanation

Feature engineering was the most important reason behind the improvement in  $R^2$  score. The original dataset contained a name column with values that included brand, model, and variant details. This column was not directly usable by Linear Regression because it was text data.

The project extracted brand and detailed vehicle model/variant information from this column. These values were then converted into dummy variables using one-hot encoding. This gave the model access to market identity information that strongly affects resale value.

```
df_model['brand'] = df_model['name'].str.split().str[0]
df_model['car_model'] = (
    df_model['name']
    .str.split()
    .str[1:4]
    .str.join('_')
)

brand_cols = [col for col in df_model.columns if col.startswith('brand_')]
model_cols = [col for col in df_model.columns if col.startswith('car_model_')]
```

This transformation helped the model distinguish between vehicles belonging to different brands and variants, which is crucial in the used-vehicle market.

## Appendix H - Model Limitations

Although the final model achieved an  $R^2$  score of 0.804, it is not perfect. Real-world vehicle prices depend on many factors that were not available in the dataset.

| Missing Factor    | Possible Effect on Price                                 |
|-------------------|----------------------------------------------------------|
| Vehicle condition | Accident-free and well-maintained vehicles sell for more |
| Service history   | Regular service records increase buyer confidence        |
| Location          | Demand and pricing differ by city and region             |
| Accident history  | Damaged vehicles lose resale value                       |

|                    |                                                |
|--------------------|------------------------------------------------|
| Insurance validity | Valid insurance can increase price             |
| Negotiation        | Actual sale price may differ from listed price |

These missing variables explain why the model does not achieve 100% accuracy. An  $R^2$  of around 0.80 is strong for a real-world used-vehicle dataset using Linear Regression.

## Appendix I - Viva Preparation

This section contains concise answers to questions that may be asked during project presentation.

| Question                           | Answer                                                                           |
|------------------------------------|----------------------------------------------------------------------------------|
| Why did you use median imputation? | Median is less affected by outliers than mean.                                   |
| Why remove duplicates?             | Duplicate records can bias analysis and inflate sample size.                     |
| Why did $R^2$ improve?             | Brand and vehicle model features captured hidden market information.             |
| Why not 100% accuracy?             | Important real-world factors such as condition and accident history are missing. |
| Which model was best?              | Model 8 with numerical, categorical, brand, and vehicle model features.          |
| What does $R^2 = 0.804$ mean?      | The model explains about 80.4% of selling price variation.                       |

The most important answer to remember is that the score improved because of feature engineering on the vehicle name column.

### Short Viva Answer

The  $R^2$  score improved to 0.804 because brand and detailed vehicle model information were extracted from the name column and encoded as additional features. These engineered features captured manufacturer-specific and model-specific price patterns, which basic numerical variables alone could not capture.

## Appendix J - Final Submission Checklist

Before submitting the project, the following checks should be completed to ensure the report and notebook are consistent.

| Checklist Item                   | Status   |
|----------------------------------|----------|
| Notebook runs from top to bottom | Required |
| All graphs visible               | Required |



|                                           |             |
|-------------------------------------------|-------------|
| Final R <sup>2</sup> score shown as 0.804 | Required    |
| M1-M6 observations included               | Required    |
| Dashboard metrics included                | Required    |
| Report PDF generated                      | Completed   |
| Backup copy saved                         | Recommended |

The project is ready for presentation once the notebook runs successfully after restarting the kernel and all outputs match the final report.

## Appendix K - Extended Code Reference

This appendix preserves the key code structure used in the project. The purpose is not to reproduce every notebook cell, but to document the main analytical steps in a readable form.

### Model Comparison Loop

```
results = []
for model_name, features in models.items():
    mae, mse, rmse, r2 = evaluate_model(features)
    results.append([
        model_name,
        len(features),
        mae,
        mse,
        rmse,
        r2
    ])

results_df = pd.DataFrame(
    results,
    columns=['Model', 'No. of Features', 'MAE', 'MSE', 'RMSE', 'R2 Score']
)

results_df = results_df.sort_values(
    by='R2 Score',
    ascending=False
).reset_index(drop=True)
```

This loop evaluated every model using the same train-test split and metrics. Keeping the evaluation process consistent ensured that the comparison between models was fair.

## Appendix L - Best Model Summary Code

The final model summary presented the best model, R<sup>2</sup> score, RMSE, and MAE in a compact table. These values were later used in the dashboard section.

```
best_model_name = results_df.iloc[0]['Model']
best_r2 = results_df.iloc[0]['R2 Score']
best_rmse = results_df.iloc[0]['RMSE']
best_mae = results_df.iloc[0]['MAE']

best_model_summary = pd.DataFrame({
    'Metric': ['Best Model', 'R2 Score', 'RMSE', 'MAE'],
    'Value': ['Model 8', f'{best_r2:.3f}', rupees_short(best_rmse),
rupees_short(best_mae)]
})
```

This summary is useful in presentations because it focuses only on the final selected model rather than all intermediate models.

## Appendix M - Dashboard Interpretation

The dashboard section summarized the entire modeling outcome in a small number of metrics. This is useful because decision-makers may not want to inspect all model details but still need to understand the final result.

| Dashboard Metric     | Meaning                                                  |
|----------------------|----------------------------------------------------------|
| Total Records        | Number of cleaned records used in the project            |
| Total Features Used  | Number of columns used by the final model after encoding |
| Best Model           | Highest-performing regression model                      |
| R <sup>2</sup> Score | Amount of price variation explained by the model         |
| RMSE                 | Prediction error with strong penalty for large errors    |
| MAE                  | Average absolute price prediction error                  |

The dashboard also helped connect the technical model evaluation to a simple business interpretation. It showed that the final model explains around 80% of price variation while maintaining a reasonable average prediction error.

## Appendix N - Final Presentation Script

This short presentation script can be used while explaining the project.

Our project is based on used-vehicle price analysis. We started by cleaning the dataset, removing duplicates, handling missing values, and converting important columns like mileage, engine, and max\_power into numerical form.

After cleaning, we performed exploratory data analysis. We found that selling prices are right-skewed and that Diesel/Petrol vehicles and Manual transmission vehicles dominate the dataset. The correlation heatmap showed that max\_power, engine, and year are important numerical predictors.

In probability analysis, we calculated the likelihood of important vehicle characteristics such as fuel type, transmission, seller type, and ownership category. In statistical inference, we estimated the average selling price and tested whether it was significantly different from Rs. 5,00,000.

For modeling, we built eight regression models. The biggest improvement came from feature engineering. We extracted brand and detailed vehicle model information from the name column and encoded it. This improved the final R<sup>2</sup> score to 0.804. This means our final model explains about 80.4% of variation in selling price.